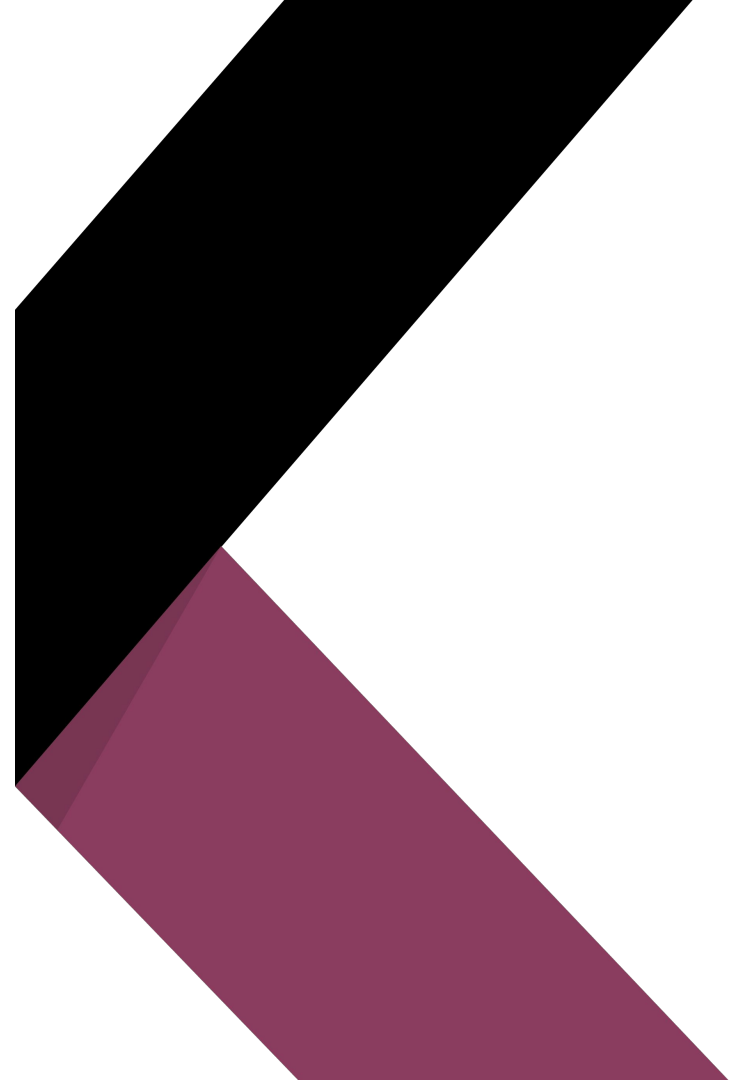


Troubleshooting Principles

Troubleshooting





Software Troubleshooting Principles

An Introduction to Troubleshooting

Troubleshooting

Troubleshooting is a form of problem solving, often applied to repair failed products or processes.



- It is a logical, systematic search for the root cause of a problem to solve it.
- Determining the most likely cause is a process of elimination, eliminating potential causes of a problem.
- Finally, troubleshooting requires confirmation that the solution restores the product or process to its working state.

“If I had an hour to solve a problem I’d spend 55 minutes thinking about the problem and 5 minutes thinking about solutions”

A. Einstein



Basic Principles Troubleshooting

Usually troubleshooting is applied to something that has suddenly stopped working.



- Initial focus is often on *recent changes* to the system or to the environment in which it exists.
- However, there is a well known principle that **correlation** does not imply **causality**.
- Therefore, troubleshooting demands **critical thinking** rather than **magical thinking**.



Critical thinking

Is the analysis of facts to form a judgement

Magical thinking

In anthropology, it denotes the attribution of causality between entities grouped with one another (coincidence) or similar to one another.



Basic Principles Troubleshooting

A basic principle in troubleshooting is to start from the simplest and most probable possible problems first.

This means to use the **Occam's razor**.



OCCAM'S RAZOR

"WHEN FACED WITH TWO POSSIBLE EXPLANATIONS, THE SIMPLER OF THE TWO IS THE ONE MOST LIKELY TO BE TRUE."

Occam's razor is the problem-solving principle that states "Entities should not be multiplied without necessity." The idea is attributed to English Franciscan friar William of Ockham (c. 1287–1347), a scholastic philosopher and theologian. It is sometimes paraphrased by a statement like "**The simplest solution is most likely the right one.**"



Basic Principles

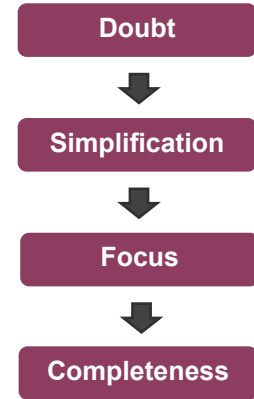
Troubleshooting

Define the Issue

- an incorrect assessment could affect all subsequent activity

○ Cartesio's Method:

- Never accept anything as true if you don't have specific evidence (Doubt)
 - *This rule helps you not to fall victim to solving the problem "on the spot": no matter how tempting many of us are, you should never jump to conclusions!*
- Divide the problem into easy-to-solve problems (Simplification)
 - *Reducing the problem into smaller parts allows you to narrow down the individual difficulties and find a solution consistent with the general problem for each individual aspect.*
- Start solving the simpler things and then move on to the more complex ones (Focus)
 - *We focus on the aspects that are easier to solve, leaving the most difficult ones for last*
- Do not omit anything from your considerations (Completeness)
 - *critically examines all aspects and related sub-problems*





Steps Troubleshooting

1. Identify the problem.
2. Establish a theory of probable cause.
3. Test the theory to determine cause.
4. If the theory was wrong discard it and go back to point 2
5. Establish a plan of action to resolve the problem and implement the solution.
6. Verify full system functionality and if applicable implement preventive measures.





- Base your theory on **facts** as much as possible.
- A theory is good if you can **test** it, even if it's the wrong one.
- **Don't be afraid of being wrong**: it is normal during the process and the environment around you should accept it.
- If you are wrong simply discard your theory and think about another.
- If you are not able to get the solution by yourself, then **help someone else** to get it: it is not important who will find the solution but to get the solution itself in a reasonable amount of time. Always share your ideas with people involved into the same issue.



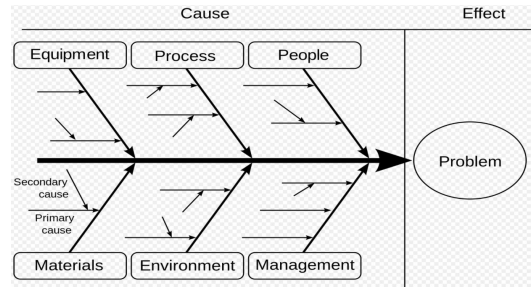
- If someone else found the solution be sure to have fully understood how it works: don't be afraid of making questions when you don't understand something. There are no silly questions: if you didn't understand, ask!
- Carefully examine available data: don't wait for “perfect” data because all you need to get a solution is often already present into first data you received, it's just a matter of being able to read them.
- Troubleshooting is a serious work but it can also be fun: you have to feel involved and interested into the problem to get good results.



Specific advices Troubleshooting

- An issue could require testing at HIL or on vehicle, if possible be present during the test to be sure that they are correctly executed: you need to trust your data.
- Have a good knowledge of system architecture: if you don't know how it works it's unlikely that you'll know how to fix it.

“When you have eliminated the impossible, whatever remains, however improbable, must be the truth.” (A.C. Doyle)



Ishikawa diagram



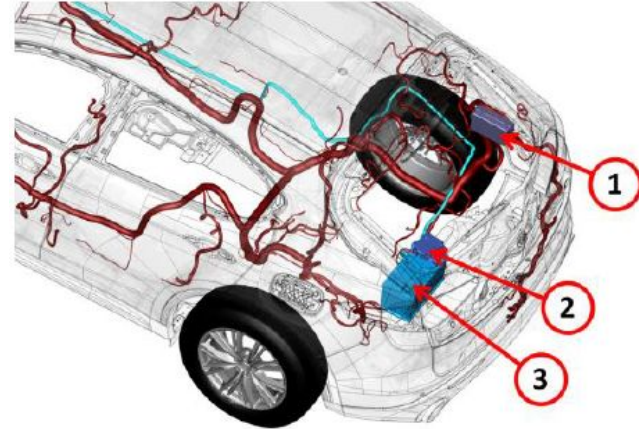
Sample Case1 Troubleshooting

CAUSES OF BATTERY DISCHARGE

If the battery is drained, its work is adversely affected.

However, it's much worse if the situation occurs repeatedly. In this case, it is worth considering why your “iron horse” behaves in such a way. In fact, some of the major causes are as follows:

- wearing of the battery;
- mismatch of the charge / discharge ratio when charging from the alternator;
- alternator failure;
- starter malfunction / poor operation;
- external leakage currents.



1. Front PDC
2. Battery Fuse Box
3. Battery



Sample Case1 Troubleshooting

Measuring Leakage Current

Before you start testing, you need to undertake preparatory work.

First of all, leave the hood open and turn off all current consumers, e.g., a car radio, exterior and interior lights, then remove the key from the ignition lock and close the door. Please remember that when you measure with a multimeter, the battery will be switching on and off, so the central lock may be triggered. It is therefore better to leave the windows open to secure access to the car.

To make measurements, we'll need:

- a multimeter with a DC measuring range not less than 10 A
- Alligator test clips (for more convenient connection);
- an 8 or 10 box or open end wrench (needs to be picked for a vehicle);
- work gloves.

It is quite convenient to measure leakage current clamp meters – no need to disconnect anything, just crimp the wire and make measurements. Clamp meters' disadvantage is that they are not too precise and can capture parasitic currents. Still, by resetting with the "Zero" button you can achieve accurate and precise results.

You need to crimp either a positive or a negative cable with all the wires connected to one of the terminals (if any). The only important point is that clamp meters should measure DC current. As a rule, their price is much higher if compared to regular clamp meters that measure only AC.





Sample Case1 Troubleshooting

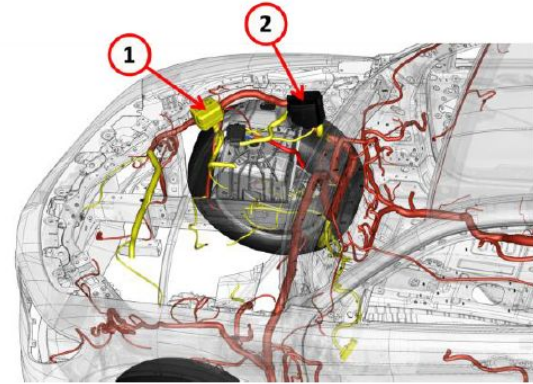
Checking Relays and Fuses

The fuse service box and relay box are located under the hood. It is also possible to place another block (box) in the passenger compartment near the dashboard, under the rear seat, and also in the boot. We can search for a potential consumer of excess current in the following way:

- connect the multimeter in the same way you connect it when measuring leakage current;
- remove each fuse one after another and insert them back in place, noting whether the current value on the multimeter display changes;
- if you find a significant (yet permissible) decrease, consult the vehicle technical documentation to learn what this fuse is responsible for and proceed to the detailed testing of devices which work with the fuse.
- Imagine you've checked all fuses, but there is still leakage current.

In this case, you need to check the devices not protected by fuses. These are:

- alternator;
- starter.



1. Fuse and relay box
2. Front PDC





Sample Case2 Troubleshooting

Torque Security Check Fail

P061B Happened on P1P4 vehicle

Customer Impact:

Contactor Open

Vehicle shutdown. It could restart after Key Cycle

Description:

PIM detecting an error in internal torque calculation:

Output torque (Estimation of Front and Rear Torque) command is outside its calculated minimum and maximum limits

Driver torque request from ECS is less than 0 at low vehicle speed

Case study:

Vehicle on Rolls in Melfi Plant

Action Required:

Reappear at HIL





Sample Case2 Troubleshooting

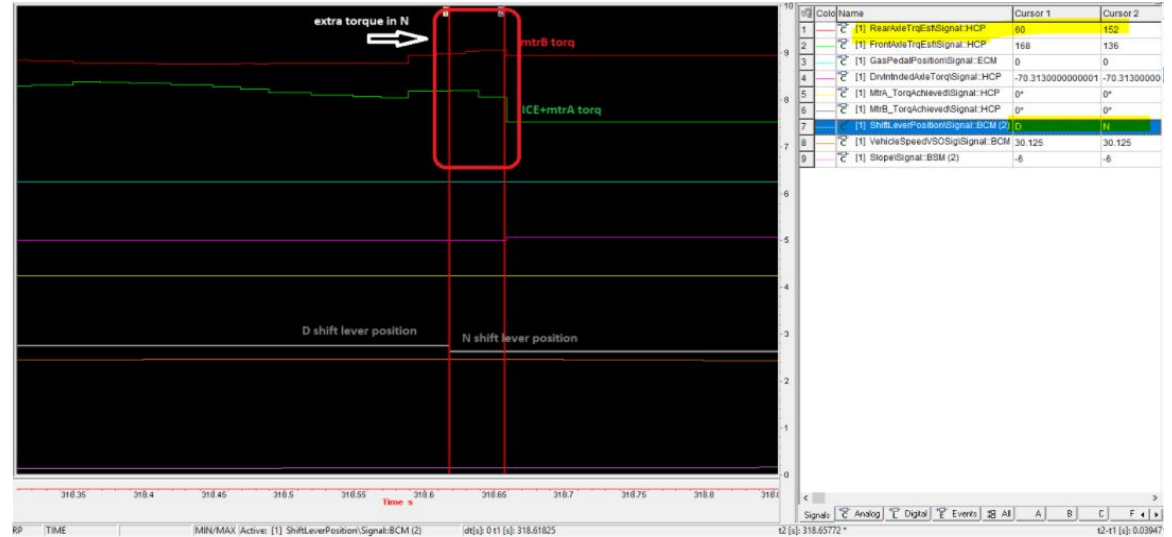
Torque Security Check Fail

I analysis

Vehicle is moving on Rolls at vehicle speed around 20 Km/h

Hybrid Mode

Fail is due to MtrB Torque (P4) happens during transition from D to N





Sample Case2 Troubleshooting

Torque Security Check Fail

II analysis

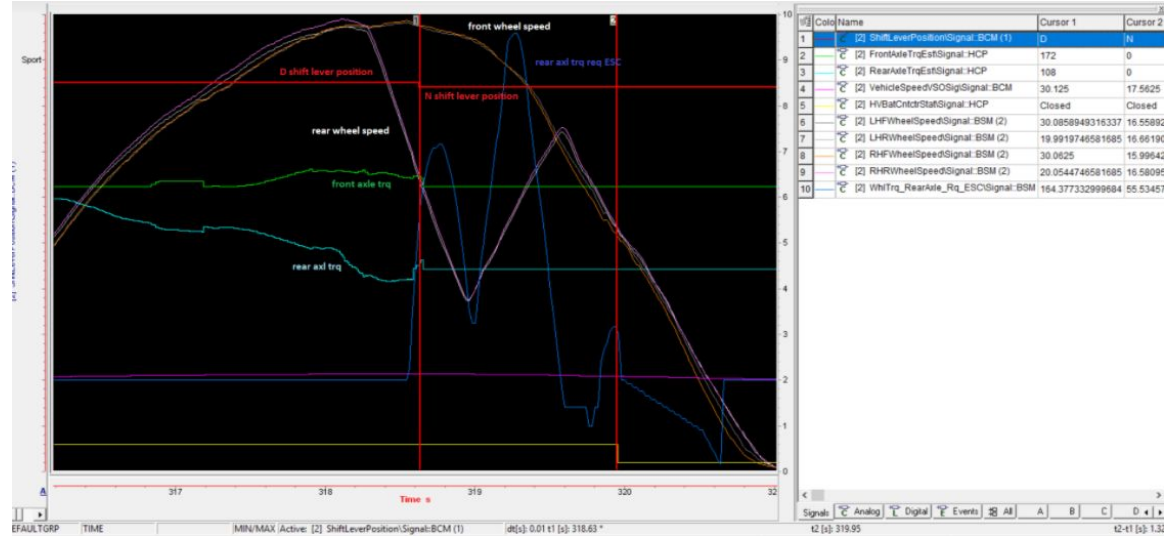
The wheels of the vehicle report a speed delta equal to 12 Km \ h with ESC intervention

Front Wheels $\approx$ 30 Km\h

Rear Wheels $\approx$ 18 Km\h

and in this moment there is also gear change from D to N

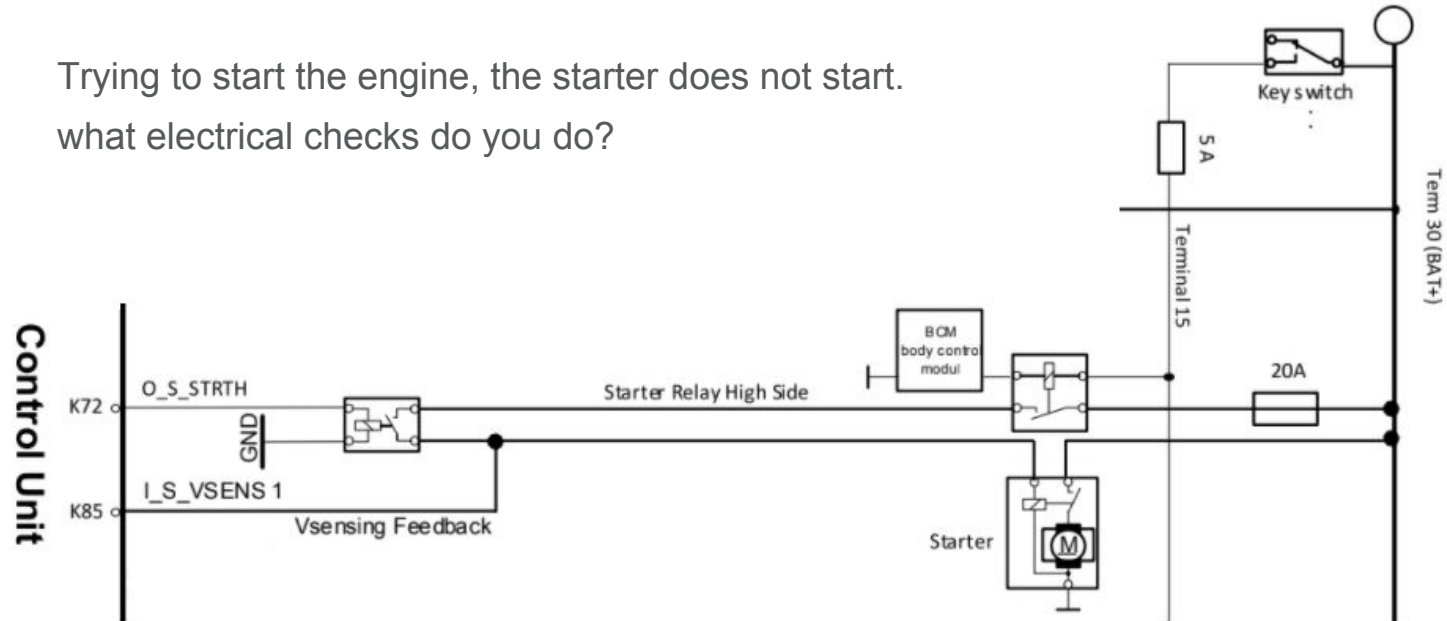
This scenario collide into the MtrB that increase and the Output Torque that overcome the Max Threshold





Exercise1 Starter

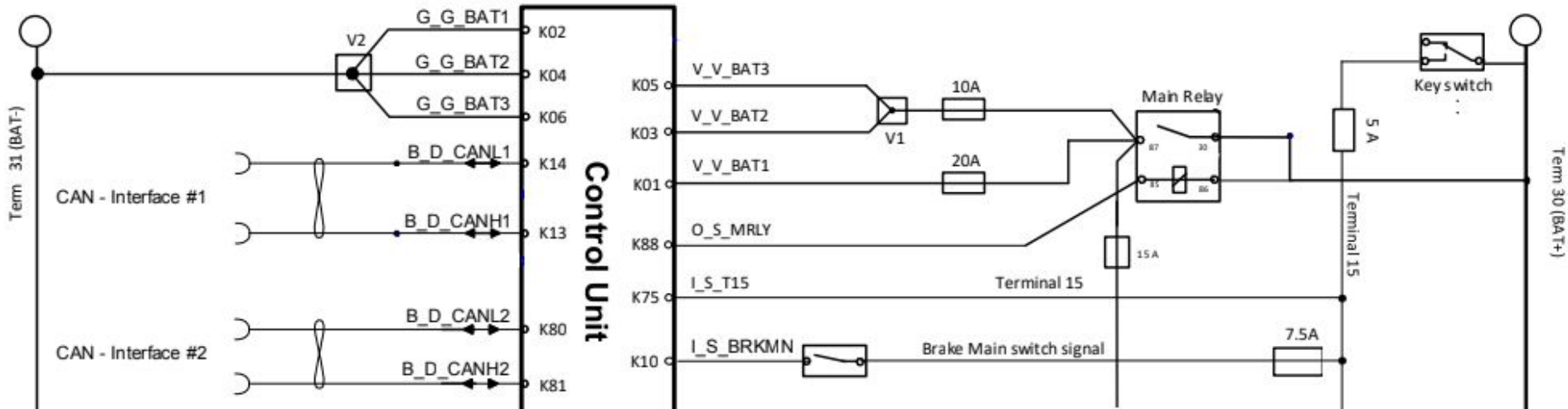
Trying to start the engine, the starter does not start.
what electrical checks do you do?





Exercise2 Node absent

in Key on conditions the ECM does not communicate on C1. What do you check?





Software Troubleshooting Principles

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